

SENNDIK®

SDA25 SERIES

SINGLE-PHASE FAULT DIAGNOSIS MODULE

A SOLID-STATE RELAY WITH INTEGRATED ERROR DIAGNOSIS FUNCTION



A SOLID-STATE RELAY WITH INTEGRATED ERROR DIAGNOSIS FUNCTION

REQUIREMENT

- ▶ 1 relay for 1 heating element+1 detection

SOLUTION

- ▶ SDA25

Why choose diagnostic relay **SHENNDIK**?
We provide various solutions for diagnostic relays. These relays can assist users in identifying faults, resistive loads, relay outputs, reducing costs and increasing efficiency.

ADVANTAGE

- ▶ These relays allow users to understand the power circuit load (connected or disconnected), relay output (closed or not closed), and protection status (circuit breaker status) through normally closed (NF) diagnostic contacts
- ▶ No potential limitation
- ▶ Single PLC input, can also be connected in series
- ▶ Easy to use
- ▶ Short reaction time<100m's

The SDA25 series intelligent solid-state relay can monitor the system status in real time during operation, assist in diagnosing the cause of faults, and provide fault alarms when the main circuit or load is open or the solid-state relay is short circuited. The fault alarm signal includes a green LED status indicator light and a cascading OD (open drain) door status output.

The input range of intelligent solid-state relays is 7~30VDC, which can better adapt to various control requirements

This product has the following characteristics:

The status light indicates the working condition of the system

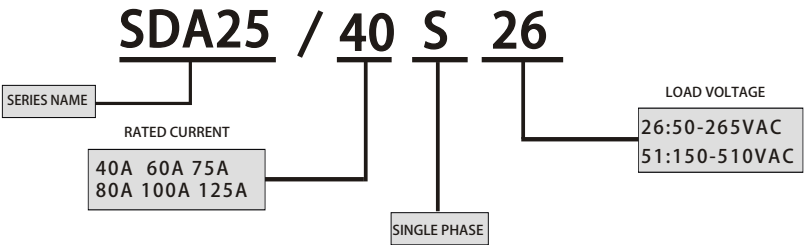
Concatenable state output

Normal state closed

Abnormal state disconnected

The green light is on normally

Abnormal green light off status



- If the main circuit voltage is below 150VAC, the solid-state relay can be used correctly, but the alarm state is no longer valid.
- When the connection between the solid-state relay and the load is relatively long (greater than 30 meters), it is not possible to detect the open circuit of the load.

CONTROL CHARACTERISTICS

Input parameters (TA=25°)	
Control voltage range	7-30VDC
Connected voltage	4VDC
Turn off voltage	1VDC
Maximum input current	28mA (@30VDC)

Status output characteristics (TA=25° C)	
output voltage	12-24VDC
output current	0.1A
Off state resistor	2Ω
Maximum cable length for open circuit load	30m
Open circuit delay (fault)	70ms
Closing delay (voltage fault)	20ms
Closing delay (current fault)	20ms

OTHER PARAMETERS

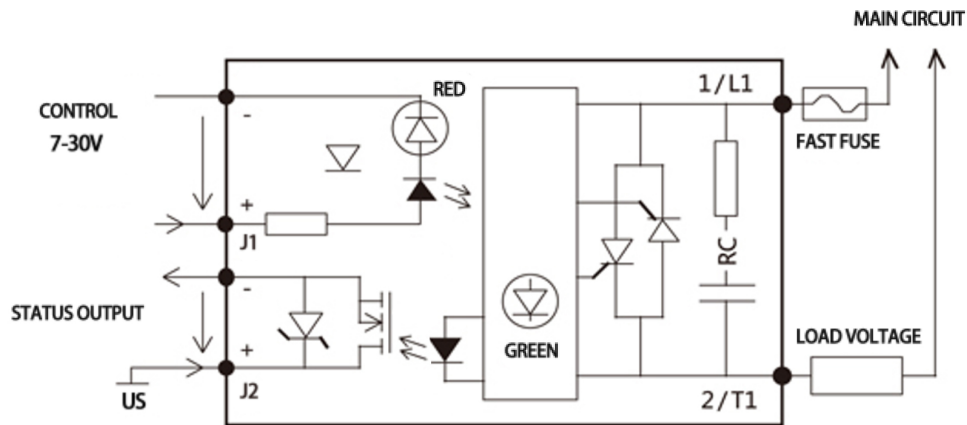
CONTROL SIGNAL (ON/OFF)	INPUT LED (RED)	INPUT LED (GREEN)	STATUS OUTPUT	LOAD POWER SUPPLY	LOAD	SOLID STATE RELAY
0			OPEN	YES	OK	OK
1			OPEN	YES	OK	OK
0			CLOSE	NO	-	-
1			CLOSE	NO	-	-
0			CLOSE	YES	OPEN	OK
1			CLOSE	YES	OPEN	OK
0			CLOSE	YES	OK	SHORT
1			CLOSE	YES	OK	SHORT

In the fault free state, the status output is usually in the off state (NF)

When there is no fault, the status output is normally closed (NC). The green LED lights up when there are no faults. This means:

- There is voltage on the relay when there is no control signal.
- When there is a control signal, there is current inside the relay.
- (Static output) usually closes when there are no faults in the relay and load.
- The green LED light is on. This means there is a malfunction When there is no control, the voltage at both ends of SSR.
- Under control, there is current inside the SSR.
- In the event of a fault (relay short circuit or load disconnection),
- The status output is on, and the green LED light is off. In the event of a load (open circuit) or SSR (short circuit) fault:
- The status output is on, and the green LED light is off. In the event of a load (open circuit) or SSR (short circuit) fault:
- The green LED light is off.

CIRCUIT SCHEMATIC DIAGRAM



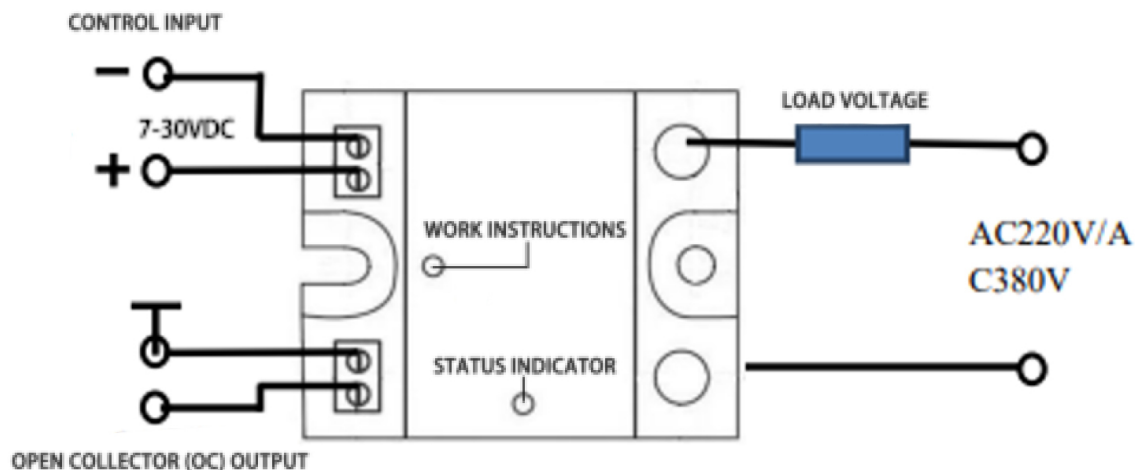
■ DURING NORMAL OPERATION, THE STATUS OUTPUT IS NORMALLY CLOSED (NC)

The brightness of green LED depends on the voltage U_E , control: on, off or load type

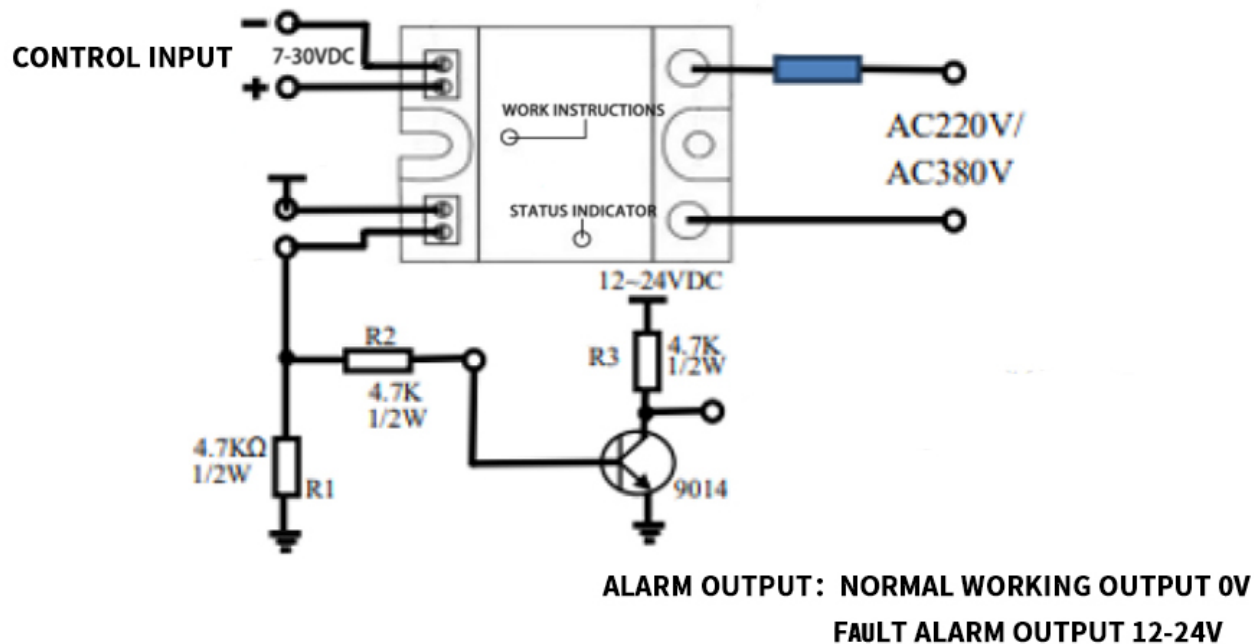
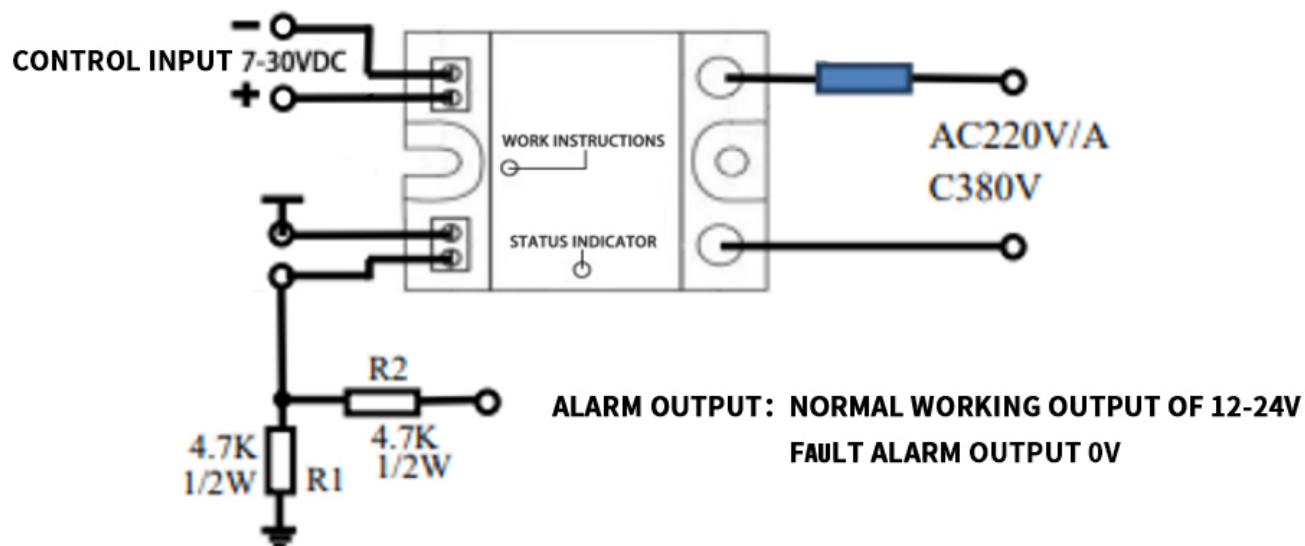
-Status output allows only NPN or PNP wiring methods

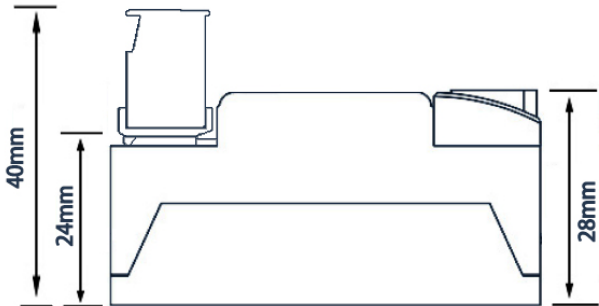
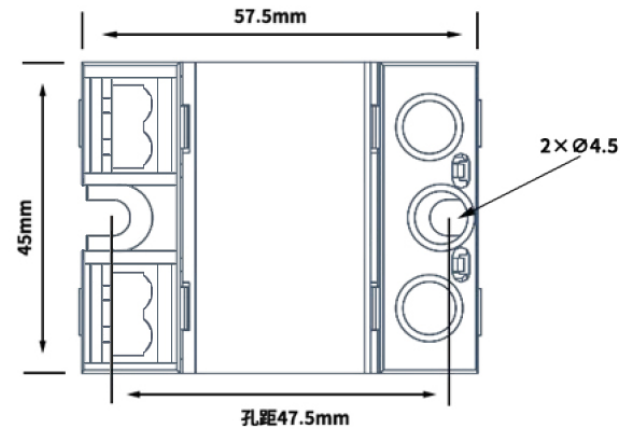
-When the voltage is below the specification (150V AC), the SSR works normally, but if the control is in the off state, it may trigger an alarm state.

WIRING DIAGRAM

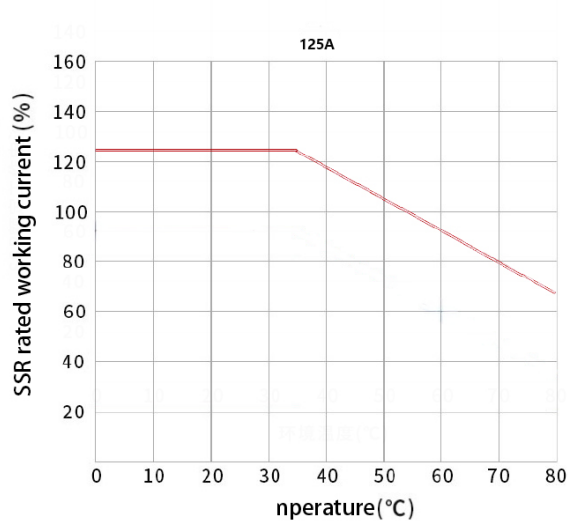
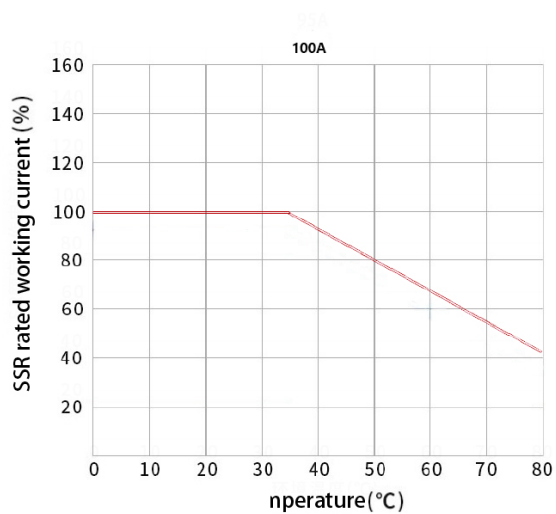
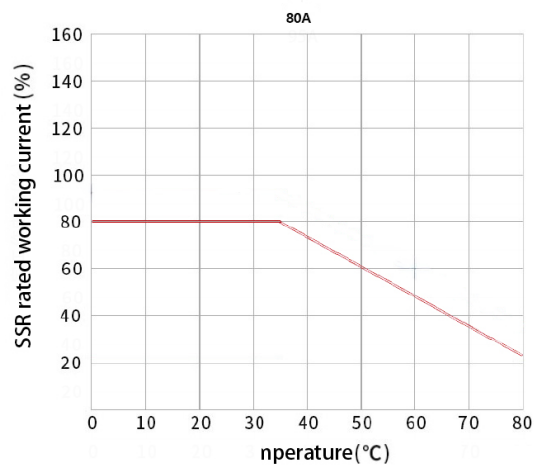
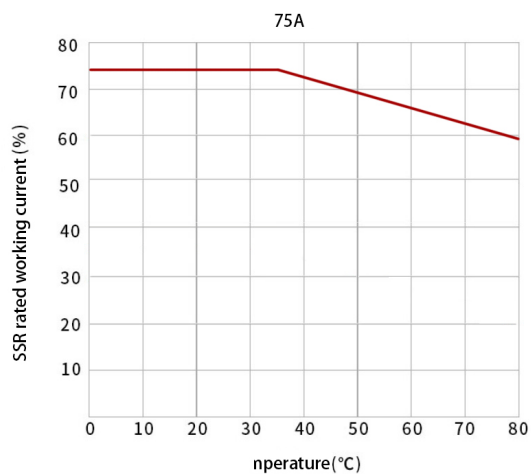
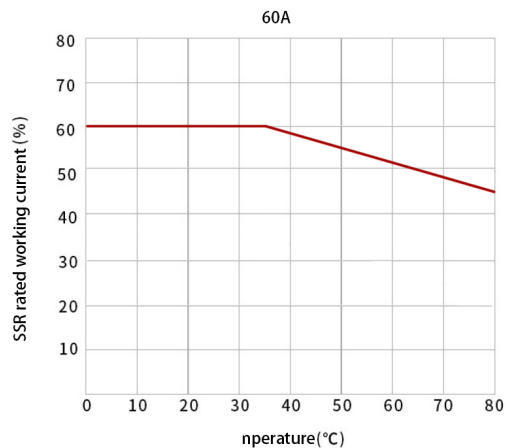
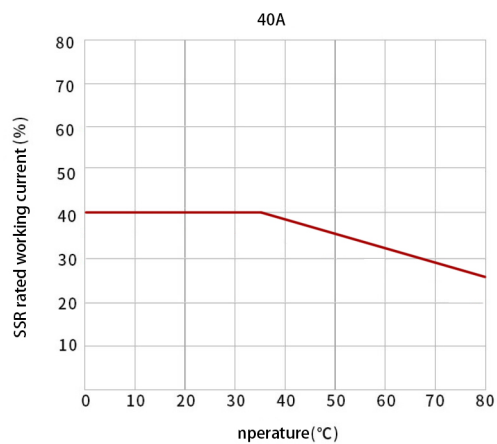


Typical usage method





TEMPERATURE



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若有更改, 恕不另行通知

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